Dhruv Saxena

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Highly motivated and detail-oriented computer science student with a strong passion for Artificial Intelligence (AI) and Machine Learning (ML). Proficient in programming languages such as Python, Java, and C++. Eager to apply theoretical knowledge in practical settings and explore the vast applications of AI and ML.

Education

Babu Banarasi Das Institute of Technology and Management , BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE — Nov 2022 - June 2026

- GPA: 8.16/10

- **Coursework:** ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING, SOFTWARE ENGINEERING, COMPUTER ARCHITECTURE

Projects

Hope Hub – AI-Powered Cancer Detection Platform

- Multi-Type Cancer Detection: Detects
- Image Upload & Prediction: Upload a .jpg or .png image and get predictions from trained deep learning models.
- PDF Report Generation: Automatically generate a detailed, downloadable PDF diagnosis report including your name, age, cancer type, and prediction.
- User Authentication: Secure signup/login system with SQLite3 database.

DQN Self-Driving Car Simulator

- A 2D self-driving car simulator powered by Deep Q-Learning (DQN). The car learns to navigate a custom race track using raycasting sensors, a 10-dimensional state vector, and reward shaping to complete laps efficiently while avoiding collisions.
- 2D Track Environment with static walls and checkpoints
- Discrete Action Space: accelerate, brake, turn left, turn right, coast
- Reward Shaping for progress, collisions, and lap completion
- Deep Q-Network (DQN) with:
  - Epsilon-greedy exploration (decay from 1.0 → 0.1)
  - Replay buffer
  - Target network updates

MOODECHO — Emotion Detection from Speech

- Real-time voice recording using audiorec in the browser.
- Automatic emotion detection using a trained CNN model on MFCC audio features.
- Confidence scores for all emotion classes.
- Lightweight model (only $\widetilde{2}$MB), ensuring fast and responsive performance.

Publications

"Beyond CNNs: A Study on Fisher Vectors and Their Fusion for Few-Shot Generalization," **International Journal of Open Information Technologies (INJOIT), Moscow University Journal, Vol. 13, No. 11, listed on Google Scholar.**

Dhruv Saxena*, Aditi Sharma*

Regularization-Driven Generalization in Few-Shot Learning: PAC-Bayes Bound as an Explanatory Tool

Dhruv Saxena

https://doi.org/10.5281/zenodo.15698715

Estimating Redshift from Galaxy Color Indices: Ridge Regression to Deep Neural Networks

Dhruv Saxena, Kunal Singh, Syed Gufran Hussain, Chitransh Anand Srivastava

https://doi.org/10.5281/zenodo.15668875

Technologies

Languages: Python, Java, C, SQL, HTML, CSS

Technologies: Computer VIsion Frameworks & Libraries: OpenCV, Sci kit learn, NumPy, Pandas, SciPy, Matplotlib. Deep Learning Machine learning

Extracurricular

Co-General Secretary – *Karmavesh Cultural Community*
July 2025 – May 2026

- Spearheaded cultural events and community initiatives.
- Coordinated between organizing committees, sponsors, and participants to ensure smooth execution of events.

Financial Head – *Voxen Gaming Community*
Aug 2025 – Jan 2026

Wrote a Novel and Published on Amazon KDP

Hobbies and Interest

Artificial Intelligence & Quantum Computing

Physics

Creative Writing & Storytelling

Leadership & Event Management

Film Making & Scriptwriting

Reading